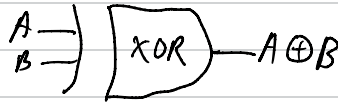


Absolute difference

① Absolute difference of binary number
XOR gate

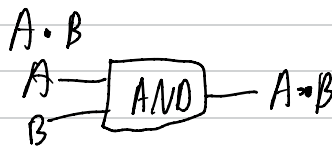


A	B	$A \oplus B$
T	T	0
T	F	1
F	T	1
F	F	0

One notice that $A \oplus B = |A - B|$
when A, B is boolean

Questions: What should we do when
 A, B are integers?

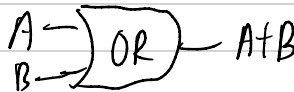
And gate



A	B	$A \cdot B$
0	0	0
1	0	0
0	1	0

OR gate

$A + B$



A	B	$A + B$
0	0	0
1	0	1
0	1	1
1	1	1

Absolute difference of multi-bit number

$$\begin{array}{r}
 6 - 1 \\
 6 \quad 0110 \\
 1 \quad 0001 \\
 \hline
 5 \quad 0111 = 7x
 \end{array}$$

We can't use XOR directly!

② Absolute difference of multibit

$|x-y| \Rightarrow$ ① Subtract
② Absolute

2. | Representation of signed integer

2's complement

We all know what is binary number
 $+34 = 100010$

but how to represent -ve number?

$+34 = 00100010$

① Flipping 11011101

② add 1 11011110

Then we can represent -ve number

Wait: $11011110 = +222$ too?

Wouldn't that confuse the computer?

★ The computer use the most significant bit (MSB) to identify +ve, -ve number

★ We need extra 1 bit to represent -ve number

2.2 Addition of multi-bit number

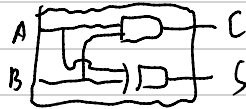
We can't do **XOR** due to the carrier (进位)

For a computer to do addition,
it must output carrier

Half-adder

$$\begin{array}{r} 01 \\ + 01 \\ \hline 10 \end{array}$$

↑ ↑
C S



A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

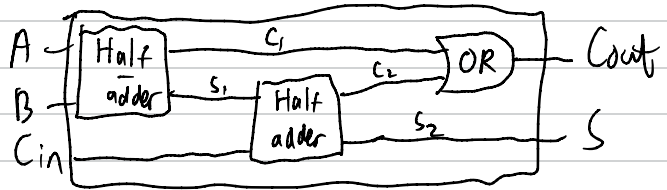
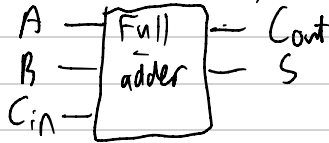
$S = A \oplus B$
 $C = A \cdot B$

Full adder

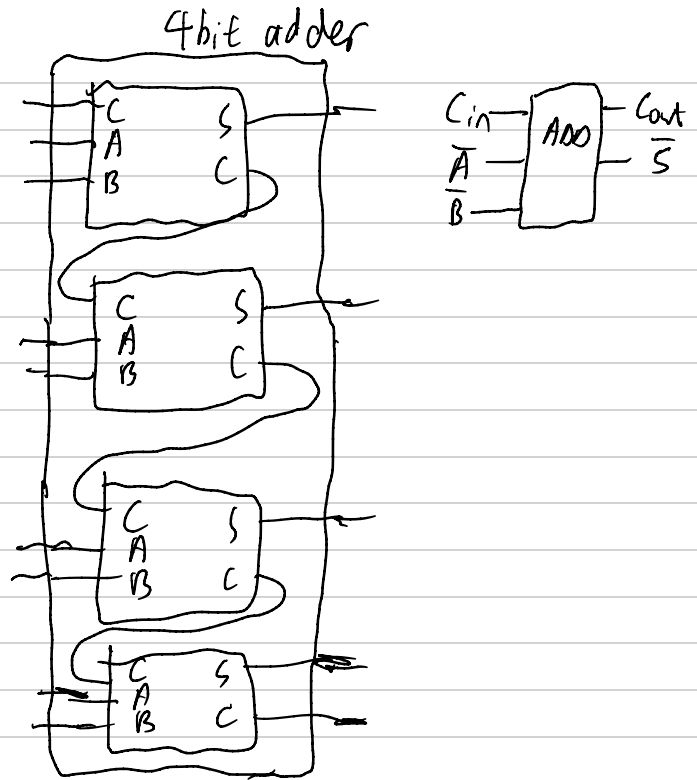
Half adder only handle the least significant bit

$$\begin{array}{r} 0001 \\ 0011 \\ \hline 1000 \end{array}$$

↑ ↑ ↑
Cout S₂ S₁



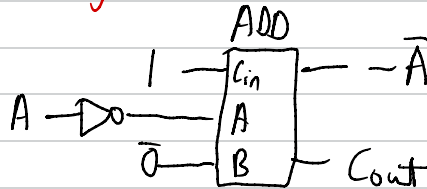
0011 Thus
0001



(2.3) Subtraction

$$X - Y = X + (-Y)$$

2's complement gate



Subtraction



② Absolute

$$\text{sign} = \text{sign}(x)$$

$$|x| = (x - \text{sign}) \oplus \text{sign}$$

$$\text{where } \text{sign}(x) = \text{MSB}(x)$$



$$\text{when } x \geq 0, \text{ sign}(x) = 0$$

$$x - \text{sign}(x) = x$$

$$x \oplus 0 = x$$

$$x < 0, \text{ sign}(x) = 1$$

$$x - \text{sign}(x) = x - 1$$

$$x - 1 \oplus 1 = -x + 1$$

③ Comparison

$$\text{Thus } |x - y| \geq \epsilon$$

$$= \text{sign}(|x - y| - \epsilon) //$$